

Master Statistics Compendium

Every number from Paper 1 in a single reference document · Cook S (국승호) & Yu H-W · 2026-05-21 · 12 rounds of analysis · 26 main figures · 25+ cohort entries

1. Cohorts & sample sizes (master inventory)

Cohort	Accession	n primary tumour	Modality	Role	Status
TCGA-THCA	thca_tcga_pan_can_atlas_2018	504	RNA-seq + HM450	Discovery + reference DM1/DM2	audit-locked
MSK-IMPACT thyroid	Landa 2016 supp	117	Targeted-panel + OS	OS replication (advanced)	audit-locked
K2 / PRJEB11591	ENA PRJEB11591	260 (RNA) / 235 (Paper-1 QC) / 34 (balanced)	RNA-seq	Korean cross-ethnic	audit-locked
Korean GSE286332	GEO GSE286332	18 (PTC 9 + PTC+HT 9)	RNA-seq FPKM	Korean ground-truth	audit-locked
Lee 2024 / GSE213647	GEO GSE213647	630 post-QC (n=632 raw)	RNA-seq	Korean independent replication	audit-locked
Mun 2025 proteogenomics	Mun 2025 supp	336 (PTC 177 + ATC 113 + PDTC 46)	MS proteomics	Cross-modality protein	audit-locked
Landa 2016 GSE76039	GEO GSE76039	37 (PDTC/ATC)	Microarray	Western advanced anchor	audit-locked
External GPL570 ×4	GSE29265 / GSE33630 / GSE53157 / GSE65144	49 + 105 + 27 + 106 = 287	Microarray	Western external direction	audit-locked
Lu 2023 single-cell	GEO GSE193581	15 patients / 67,678 cells (14,624 malignant)	sc 10x	Thyocyte-intrinsic sc	audit-locked
Pu 2021 single-cell	GEO GSE184362	15 samples / 134,407 cells (10 PTC patients)	sc 10x	Second sc cohort	audit-locked NEW 2026-05-21
DepMap 23Q4 CRISPR	depmap.org	1,141 cell lines / 11 thyroid lineage	CRISPR Chronos	Druggable target stratification	audit-locked
DepMap PRISM drug screen	depmap.org PRISM	1,518 drugs / 470-480 cell lines	Drug screen LFC	Translational hook	audit-locked

Total tumour samples (unique): 504 + 117 + 260 + 18 + 630 + 336 + 37 + 287 + 15 sc + 15 sc + 1,141 CL + 480 PRISM CL ≈ 3,840 tumours/samples + 149,000+ single cells + 1,621 cell lines tested.

2. Headline effect sizes (audit-locked)

Effect	Source	Statistic
TCGA d4p2 DM1 vs DM2 ref	n=500	Cohen's d = 1.78, $p = 7 \times 10^{-45}$, ARI 0.49
Pan-genome top-5000 ARI	TCGA-THCA	ARI = 0.92 (vs DM1/DM2)
TIERA67 67-gene ARI	TCGA-THCA	ARI = 0.90; hypergeom $p = 3 \times 10^{-4}$ vs pan-genome top-100
Driver_anchor 12 ARI	TCGA-THCA	ARI = -0.007 (random)
BRAF V600E vs WT mRNA	TCGA-THCA	Cohen's d = -0.04 NS (driver-orthogonal)
K2 within-cohort unsup KMeans	n=260	Cohen's d = 1.94, DM1 prev = 60.8% (158/260)
K2 transfer-prob ranking AUC	n=260	AUC = 0.883 vs within-K2 unsup label
K2 balanced n=34 RAI ₈ AUC	n=34	0.275 (calibration drift artefact)
Lee 2024 within-cohort unsup	n=632 all / 370 tumour	Cohen's d = 5.93 / 5.48; 7-of-8 panel genes
Mun 2025 protein unsup	n=336	Cohen's d = 2.86; ATC 95.6%, PDTC 67.4% → DM1
Landa GSE76039 unsup	n=37	Cohen's d = 4.08; 45.9% DM1
GSE286332 PTC vs PTC+HT	n=18 (Korean ground-truth)	Cohen's d = 1.60; TF ₃ AUC 0.938
Lu 2023 sc ATC vs PTC	n=14,624 malignant cells / 15 patients	Cohen's d = 3.04; 100% per-patient sign-consistent
Pu 2021 sc Tumor vs Paratumour	n=134,407 cells / 15 PTC samples	Cohen's d = 0.51 (smaller contrast)
Cross-cohort master forest	14 entries × 11 cohorts × 4 modalities	Mean d = 2.79; all 14-of-14 direction-consistent
DM1 fusion enrichment	cBioPortal SV n=542	76.8% (63/82) DM1; OR = 7.41 [4.32, 12.74]; $p < 10^{-15}$
RET-fusion capture by DM1	TCGA	27/33 = 81.8%
Driver-stratified DM1	TCGA d4p2 n=500	BRAF V600E+ 0.7% / BRAF/RAS-neg 49.1% / RAS+ 96.4%; OR 4.78 [3.15, 7.24], $p = 6.5 \times 10^{-14}$
Pooled OS HR	TCGA HR 2.30 [0.77, 6.88] + MSK HR 2.67 [1.17, 6.10]	Pooled HR 2.53 [1.31, 4.89], Cochran I ² =0%
TCGA PFI Cox PH (multivariate, honest)	n=432, 27 events	DM1 HR = 0.78 [0.18, 3.48], $p = 0.74$ NS (underpowered)
HM450 TPO methylation DM1 vs DM2	n=503	Cohen's d = 2.99, $p = 1.6 \times 10^{-11}$
HM450 mean 8-gene β DM1 vs DM2	n=503	0.385 vs 0.253 (52% relative increase); d = 2.18
Within-DM1 fusion+/- cell composition	n=74 vs 391	Cohen's d ≤ 0.42 across 8 cell-type fractions → fusion-independent
NRI TDS-16 vs RAI ₈	5-fold CV LogReg	+0.777 [+0.513, +1.040] (alternative main panel)
NRI TF ₃ vs RAI ₈	5-fold CV LogReg	-1.194 [-1.487, -0.901] (identity-only undercall)
Permutation null AUC (5,000 shuffles)	n=500	empirical $p = 0.0002$
Panel-conditional null (10,000 shuffles)	n=500	$p < 10^{-4}$ (no shuffle reaches $ d \geq 1.78$)
Thyroid-matched bio-prior null	n=12,921 pool, 1,000 draws	$p_{\text{emp}} = 0.007$ (top 0.7%)
Knock-out n-1 floor	—	min d = 1.54 (DIO1); all 8 genes > 1.5
Cluster stability bootstrap	500 iter	97.4% of tumours ≥ 90% stable assignment
Cross-method consensus	3 methods (DL REM, RE2, ComBat)	TF ₃ I ² =35.9% (best); NONOVERLAP ₈ ComBat 0.700 (best); RAI ₈ RBF-SVM 0.963
DepMap MYC essentiality (DM1-high vs DM1-low)	n=372 vs 394	$\Delta = -0.376$, Cohen's d = -0.50, $p = 1.0 \times 10^{-11}$
DepMap NAMPT	same	$\Delta = -0.194$, d = -0.45, $p = 1.5 \times 10^{-9}$
DepMap thyroid lineage (n=11)	—	MYC 100% essential, DNMT1 100% essential, PAX8/ NKX2-1 9% (lineage-TF role)
PRISM AZD-0364 (MEK inhibitor)	n=372 vs 394	Cohen's d = -0.59, FDR = 5.9×10^{-7} , $\Delta\text{LFC} = -0.94$
PRISM significant drugs at FDR<0.05	1,518 drugs tested	11 hits, all DM1-high selective; 7/11 canonical MAPK pathway
	n=1,287, 5 cohorts	$\rho = -0.327$ [-0.376, -0.278], I ² = 72.9%

Effect	Source	Statistic
MAPK $\times$ Panel-8 cross-cohort Spearman		
Single-cell intratumoral DM1 fraction	Lu 2023 malignant	PTC 10-40% silenced; ATC 80-100% silenced; per-cell std 0.374 (PTC) vs 0.18 (ATC)
Per-gene $\times$ per-cohort Cohen's d	8 genes $\times$ 10 contrasts	67-of-77 cells (87.0%) direction-consistent positive

3. Figures inventory (26 main + 2 supp)

#	Figure	Section	Key data
1	2-layer architecture (TIERA hierarchy)	§ 2.1	ARI ladder + Driver_anchor random
2	Gene-set hierarchy	§ 2.2	Pan-genome $\supset$ TIERA67 $\supset$ TDS-16 $\supset$ RAI ₈ $\supset$ TF ₃
3	RAI biology pathway (pretty)	§ 2.2	Follicular cell + apical/basolateral 3-zone
4	Cross-cohort 16-panel scan	§ 2.3	TCGA $\times$ K2 $\times$ GSE286332
5	TCGA panel scan AUC + ARI	§ 2.3	16 panels, RAI ₈ anchor
5b	Driver-stratified DM1 (NEW)	§ 2.5	OR 4.78 + biology card
5c	Biology mega-stratification (NEW)	§ 2.5	16 strata $\times$ 6 cohorts
5d	Per-gene $\times$ cohort matrix (NEW)	§ 2.5	77 cells, 87% direction-consistent
6	External direction-consistency forest	§ 2.3	GPL570 $\times$ 4 cohorts
7	NONOVERLAP zero-overlap scatter	§ 2.3	ρ -0.84 to -0.94
8	External histology boxplots	§ 2.3	ATC always lowest
9	Quantum kernel SVM (honest)	§ 2.4	RBF 0.963 vs quantum 0.369
10	Survival audit OS vs PFI	§ 2.6	Pooled HR 2.53
11	DM1 robustness forest 12 contrasts	§ 2.7	10/12 FDR<0.01
11b	Cross-cohort master forest (NEW, 14 entries)	§ 2.7	Pu 2021 added 2026-05-21
12	Per-gene heatmap	§ 2.7	TG/FOXEl/TSHR/TPO top
13	Knock-out n-1 LOO	§ 2.7	floor d=1.54
14-17	ROC/KM/Calibration/DCA quad	§ 2.6	RAI ₈ AUC 0.963 RBF-SVM
18	HM450 per-gene	§ 2.7	TPO d=2.99 lead
19	Mechanism cascade 4-step	§ 2.7	TF $\rightarrow$ DNMT $\rightarrow$ methyl $\rightarrow$ effector
20	sc + DepMap + PRISM + HM450 4-panel (NEW)	§ 2.7	sc d=3.04 + MYC + drug
21	PRISM drug screen (NEW)	§ 2.7	AZD-0364 FDR 6e-7
22	Intratumoral heterogeneity Lu 2023 (NEW)	§ 2.7	PTC 10-40% DM1 / ATC uniform
23	Pu 2021 second sc cohort (NEW 2026-05-21)	§ 2.7	134k cells, d=0.51 T vs P
S1	Combined LOO + Korean cohort applicability	Supp	RAI ₈ NONOVERLAP ₈ panel-pair
S2	Korean cohort applicability board	Supp	DM1 prevalence Korean vs TCGA

4. Multi-method consensus (3 statistical lenses)

Panel	z-mean	DL REM	RE2 I ² %	ComBat pooled	RBF-SVM	Quantum	Role
RAI ₈ canonical	0.861	0.549	95.1	0.617	0.963	0.369	MAIN
NONOVERLAP ₈	0.877	0.719	90.3	0.700	0.924	0.534	Supp A
TF ₃	0.784	0.725	35.9	0.620	0.894	0.597	Supp B
TDS-16 (Yoo)	0.882	0.660	92.9	0.649	—	—	Supp C
Mechanism arm 5	0.171*	0.465*	93.6	0.349*	—	—	Supp D

5. Author actionable items (final)

Item	Status	Notes
§ 1.1 Hook	author-keyboard	Alt B refined 39w (current) vs hybrid compass 42w
§ 3.1 Discussion opening	author-keyboard	Landa 2016 JCI 126:1052-1066 cite; 3-layer reverse-causality
§ 3.4 Limitations	author-keyboard	4 main + 2 supp limitations (current Claude draft, structure locked)
Q9 reviewer hardest	author-keyboard	TF ₃ vs RAI ₈ main panel framing b (current Claude draft)
Cover letter ¶ 1	author-keyboard	NComms editor hook (anchor facts in cover letter draft)
ORCID + 교신저자 institutional email	author info	Yu Hyeong-won @snubh.org expected pattern
45+ reference PubMed verification	external	22 current → 45+ planned; specific volume/issue verification
GitHub release tag	submission-time	paper1-ncomms-submission-2026-05-21
Zenodo DOI mint	submission-time	Archive linked to GitHub release
Bundang prospective recruitment	future work	Power analysis: n=384-430 needed for HR=2.5 detection at PFI 11.6%

Master statistics compendium compiled 2026-05-21 · Cook S (국승호) & Yu H-W · accompanies main bioRxiv preprint + NComms submission package + Methods v2 + Reviewer defense matrix + Supplementary data bundle.